

Question 1:

- a) $U(m) = m$.
- i) A1: \$5,
 - ii) A2: $0.6*8 + 0.4*0 = \$4.8$
 - iii) B1: $\frac{1}{3}*0 + \frac{1}{3}*6 + \frac{1}{3}*12 = \6
 - iv) B2: \$5
- b) Pacman would receive ticket B2, since the best he'll get between 4.8 (A2) and 5 (B2) is B2.
- c) $U(m) = m^2$
- i) A1: $1*5^2 = 25$
 - ii) A2: $0.6*8^2 + 0.4*0^2 = 64$
 - iii) B1: $\frac{1}{3}*0^2 + \frac{1}{3}*6^2 + \frac{1}{3}*12^2 = 12 + 48 = 60$
 - iv) B2: $1*5^2 = 25$
- d) Ghost would provide select tickets A2 and B2 for pacman, thinking $U(m) = m$. However, this is to pacman's benefit, as he would choose ticket type A and get 64 as a benefit (A2).

Question 2:

- a) See Answer Sheet
- b) Q's Calcs
- i) $Q(A1, Up) = 0.15*(1 + 0.9V*(A1)) + 0.75*(1 + 0.9V*(B1)) + 0.10(1 + 0.9V*(A2))$
 - ii) $Q(A1, Down) = 0.85*(1 + 0.9V*(A1)) + 0.05*(1 + 0.9V*(B1)) + 0.10(1 + 0.9V*(A2))$
 - iii) $Q(A1, Left) = 0.85*(1 + 0.9V*(A1)) + 0.10*(1 + 0.9V*(B1)) + 0.05(1 + 0.9V*(A2))$
 - iv) $Q(A1, Right) = 0.15*(1 + 0.9V*(A1)) + 0.10*(1 + 0.9V*(B1)) + 0.75(1 + 0.9V*(A2))$
 - v) $Q(A2, Up) = 0.10*(5 + 0.9V*(A1)) + 0.15*(5 + 0.9V*(A2)) + 0.75(5 + 0.9V*(B2))$
 - vi) $Q(A2, Down) = 0.10*(5 + 0.9V*(A1)) + 0.85*(5 + 0.9V*(A2)) + 0.05(5 + 0.9V*(B2))$
 - vii) $Q(A2, Left) = 0.75*(5 + 0.9V*(A1)) + 0.15*(5 + 0.9V*(A2)) + 0.10(5 + 0.9V*(B2))$
 - viii) $Q(A2, Right) = 0.05*(5 + 0.9V*(A1)) + 0.85*(5 + 0.9V*(A2)) + 0.10(5 + 0.9V*(B2))$
- c) Q's calcs:
- i) $Q(A1, Up) = 0.15*(1 + 0.5V*(A1)) + 0.75*(1 + 0.5V*(B1)) + 0.10(1 + 0.5V*(A2))$
 - ii) $Q(A1, Down) = 0.85*(1 + 0.5V*(A1)) + 0.05*(1 + 0.5V*(B1)) + 0.10(1 + 0.5V*(A2))$
 - iii) $Q(A1, Left) = 0.85*(1 + 0.5V*(A1)) + 0.10*(1 + 0.5V*(B1)) + 0.05(1 + 0.5V*(A2))$
 - iv) $Q(A1, Right) = 0.15*(1 + 0.5V*(A1)) + 0.10*(1 + 0.5V*(B1)) + 0.75(1 + 0.5V*(A2))$
 - v) $Q(A2, Up) = 0.10*(5 + 0.5V*(A1)) + 0.15*(5 + 0.5V*(A2)) + 0.75(5 + 0.5V*(B2))$
 - vi) $Q(A2, Down) = 0.10*(5 + 0.5V*(A1)) + 0.85*(5 + 0.5V*(A2)) + 0.05(5 + 0.5V*(B2))$
 - vii) $Q(A2, Left) = 0.75*(5 + 0.5V*(A1)) + 0.15*(5 + 0.5V*(A2)) + 0.10(5 + 0.5V*(B2))$
 - viii) $Q(A2, Right) = 0.05*(5 + 0.5V*(A1)) + 0.85*(5 + 0.5V*(A2)) + 0.10(5 + 0.5V*(B2))$

Question 5:

- a) New $V(s) = (\text{old } V(s)) + \alpha * (\text{immediate_reward_of_state} + \text{discount} * \text{value_of_next_state} - (\text{old } V(s)))$
- i) Since T is terminal, state that $V(T) = 10$
 - ii) $V(B1) = 0 + 0.1(0 + 0.5*0 - 0) = 0$
 - iii) $V(A1) = 0 + 0.1(0 + 0.5*0 - 0) = 0$
 - iv) Same until $V(B4) = 0 + 0.1(0 + 0.5*0 - 0) = 0$
 - v) $V(B5) = 0 + 0.1(10 + 0.5*0 - 0) = 1$
- b) Trial 2:
- i) $V(B1) = 0 + 0.1(0 + 0.5*0 - 0)$
 - ii) $V(B2) = 0$
 - iii) $V(C2) = 0 + 0.1(2 + 0.5*0 - 0) = 0.2$
 - iv) $V(C3), V(B3) = 0,$
 - v) $V(B4) = 0 + 0.1(0 + 0.5*1 - 0) = 0.05$
 - vi) $V(B5) = 1 + 0.1(10 + 0.5*0 - 1) = 1.9$
- c) Trial 3:
- i) $V(B1) = 0 + 0.1(0 + 0.5*0 - 0)$
 - ii) $V(C1) = 0 + 0.1(0 + 0.5*0.2 - 0) = 0.01$
 - iii) $V(C2) = 0.2 + 0.1(2 + 0.5*0 - 0.2) = 0.38$
 - iv) $V(C3) = 0 + 0.1(0 + 0.5*0 - 0) = 0$
 - v) $V(C4) = 0 + 0.1(0 + 0.5*0.05 - 0) = 0.0025$
 - vi) $V(B4) = 0.05 + 0.1(0 + 0.5*1.9 - 0.05) = 0.14$
 - vii) $V(B5) = 1.9 + 0.1(10 + 0.5*0 - 1.9) = 2.71$
- d)